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Introduction 1

This project focused on developing a machine learning model to predict credit card default risk. The primary objective was to identify applicants who are likely to default on their credit card payments.

1.1 Dataset Overview

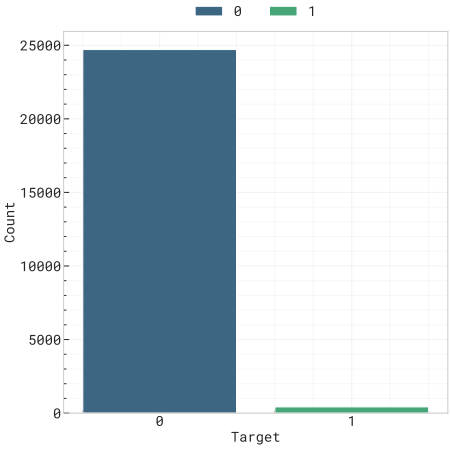

Figure 1.1: Counts of Target Variable

The dataset consisted of 25,134 applicant records with 17 features, including variables such as gender, car ownership, housing status, employment details, income, and education. The target variable, labeled TARGET, indicated whether an applicant defaulted (1) or not (0). A key challenge identified early in the analysis was the severe class imbalance in the dataset.

Non-defaulters made up 98.32% of the data (24,712 records), while defaulters accounted for only 1.68% (422 records), resulting in an imbalance ratio of approximately 58:1. This imbalance significantly influenced model design and evaluation strategies.

1.2 Data Cleaning

Data cleaning involved multiple steps to ensure data quality. Unrealistic age values above 100 were removed. Missing values in YEARS_EMPLOYED and FAMILY_SIZE were handled using median imputation, while INCOME_TYPE was imputed using the mode. The FLAG_MOBIL feature was dropped because it contained no variance and therefore provided no predictive value.

Exploratory Data Analysis 2

2.1 Univariate Analysis

The distribution of the two continuous variables, income and age, are shown in Figure 2.1 and Figure 2.2 respectively. The income distribution is heavy skewed to the right, with a median income of 2610000. The age distribution is approximately normal, with a mean age of 43.2 years. Therefore the choice of median income as a imputing value is justified.

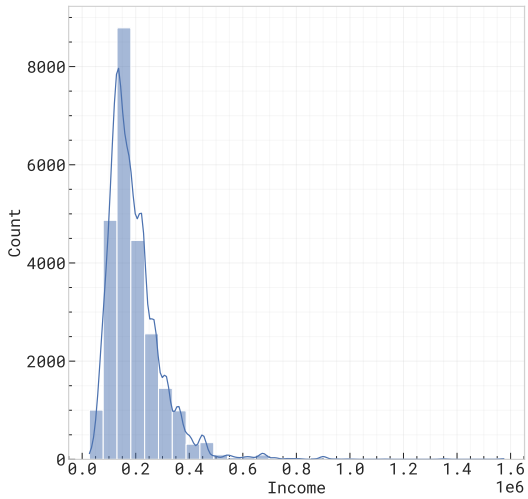

Figure 2.1: Income Distribution

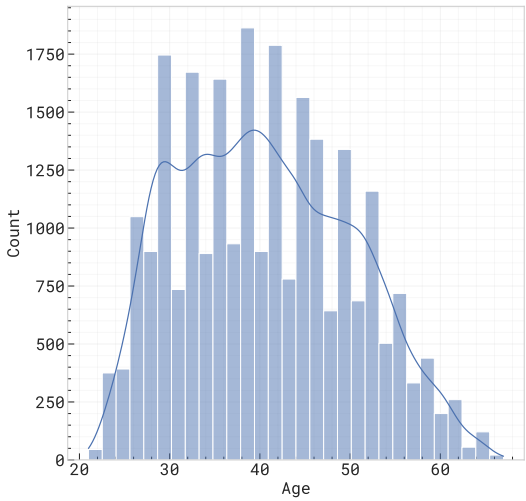

Figure 2.2: Age Distribution

2.2 Bivariate Analysis

Looking at the relationship between income and age, we cannot see that correlation between income and age.

This would be natural as the income of a person is not directly related to their age. It could vary for many reasons such as

- Type of employment
- Level of education
- Experience

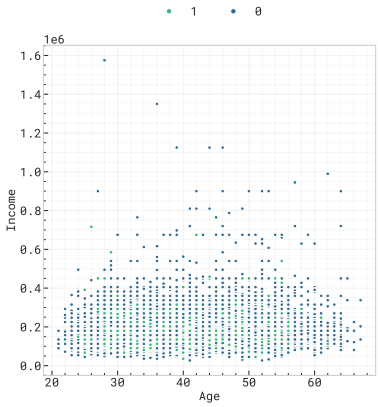

Figure 2.3: Income vs Age Distribution

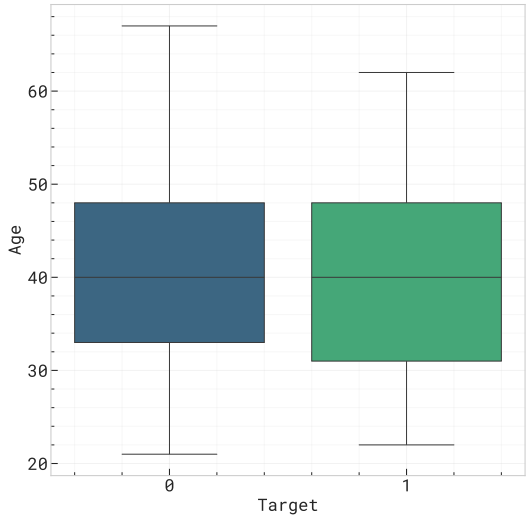

Figure 2.4: Age Box Plot by Target

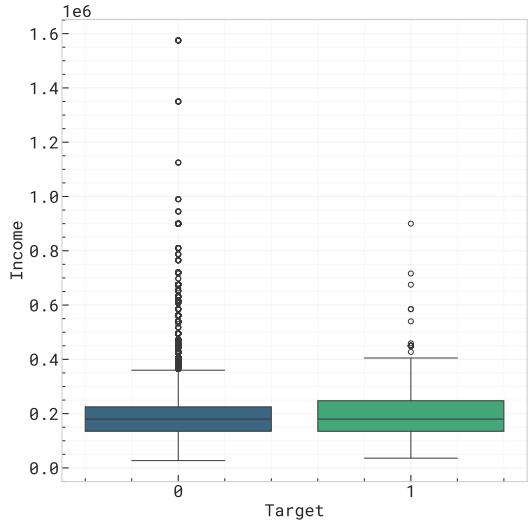

Figure 2.5: Income Box Plot by Target

The two boxplots above show that there is no significant diference in the quartiles of the age and the income between defaulters and non-defaulters.

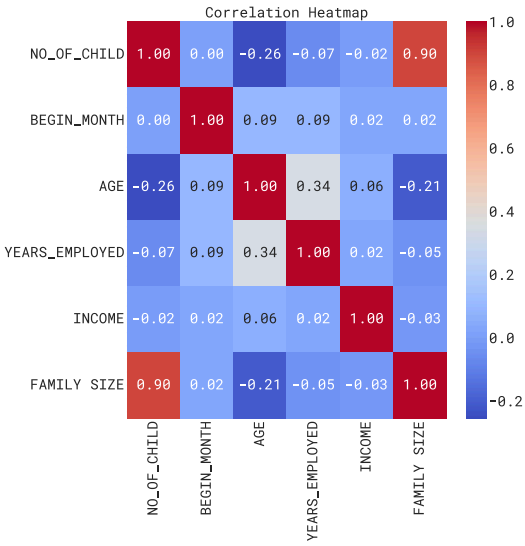

Figure 2.6: Correlation Matrix

The correlation matrix shows the correlation between all the numeric features in the dataset. Which is important identify if there is a possibilty of having interactions amoung predictor varibles. Which would be important for ordinary statistical models. However, for machine learning models such as random forests, it is not that important as they can capture interactions.

Advanced Data Analysis 3

3.1 Initail models

3.1.1 Logistic Regression

The Logistic Regression is the simplest model for a binary classification problem. It is a linear model that uses a logistic function to map the output of a linear equation to a probability between 0 and 1. The results are as follows:

```
LogReg
   CV F1 : 0.0288 ± 0.0257
   Test  — Precision: 1.0000  Recall: 0.0357  F1: 0.0690  AUC: 0.7178
   TP=3  TN=4,942  FP=0  FN=81
```

Figure 3.1: Logistic Regression Result

It can be seen that the model is not able to predict the positive class well. This is expected as the dataset is highly imbalanced.

3.2 Balanced Logistic Regression

```
LogReg_balanced
   CV F1 : 0.0521 ± 0.0005
   Test  — Precision: 0.0301  Recall: 0.6667  F1: 0.0576  AUC: 0.7116
   TP=56  TN=3,138  FP=1,804  FN=28
```

Figure 3.2: Balanced Logistic Regression Result

By Balancing the dataset, by weighting the minority class we can see that the model is able to predict the positive class significantly well. To see if there could be an improvement Machine Learning Models were trained.

3.3 Machine Learning Models

A few Machine Learning Models including:

- RandomForest
- GradientBoosting
- XGBoost

including thier balanced versions were trained. The results are as follows:

MODEL COMPARISON

	Model	CV F1 Mean	CV F1 Std	Test F1	Precision	Recall	ROC-AUC	TP	TN	FP	FN
2	RandomForest	0.2378	0.0751	0.2281	0.4333	0.1548	0.7841	13	4,925	17	71
3	RandomForest_balanced	0.2052	0.0674	0.2478	0.4828	0.1667	0.7688	14	4,927	15	70
6	XGBoost_weighted	0.1591	0.0218	0.1364	0.0799	0.4643	0.7678	39	4,493	449	45
5	XGBoost	0.1585	0.0456	0.1800	0.5625	0.1071	0.7510	9	4,935	7	75
4	GradientBoosting	0.1013	0.0288	0.1111	0.8333	0.0595	0.7559	5	4,941	1	79
1	LogReg_balanced	0.0521	0.0005	0.0576	0.0301	0.6667	0.7116	56	3,138	1,804	28
0	LogReg	0.0288	0.0257	0.0690	1.0000	0.0357	0.7178	3	4,942	0	81

Figure 3.3: Model Comparison

By looking at the table of model comparisons it can be seen that the the RandomForest and Balanced RandomForest models seem to have higher Test F1 scores and ROC-AUC values they are bad at predicting defaulters accurately. Same goes for XGBoost. However GradientBoosting seems to be better at Precision as in being being able to reduce the number of false negative is able to predict the positive class significantly well. Eventhough the Logistic Regression (Balanced) has a slighly lower Test F1 Score and Precision it has the best Recall and comparatively similar ROC-AUC measure as well, which in this case would make it the best model out of the set of models.

3.3.1 Balanced Machine Learning Models

To see if improvements could be made

3.3.2 Model Stacking

3.4 Summary

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